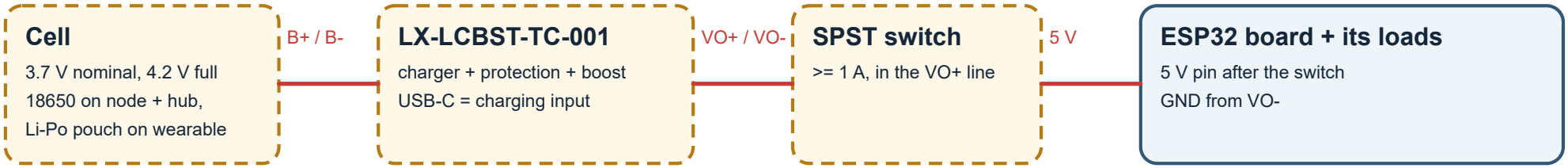
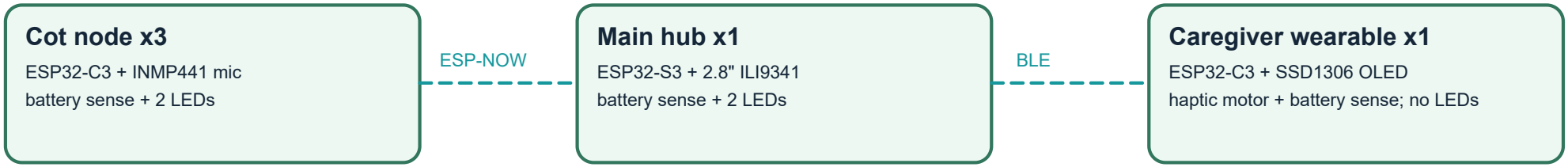
1 / CryNet system - power chain and links

Every unit has its own battery and its own ground. No wire ever joins two units.



One chain in all four units - only the loads differ:



Rules every unit follows:

1 - Switch OFF before charging. The cell still charges on B+/B-.

2 - Set the boost trimpot to 5.1 V with a meter before any board.

3 - Never power a board from USB and the boost at the same time.

Dashed teal = wireless link, no wire. Solid = a real conductor.

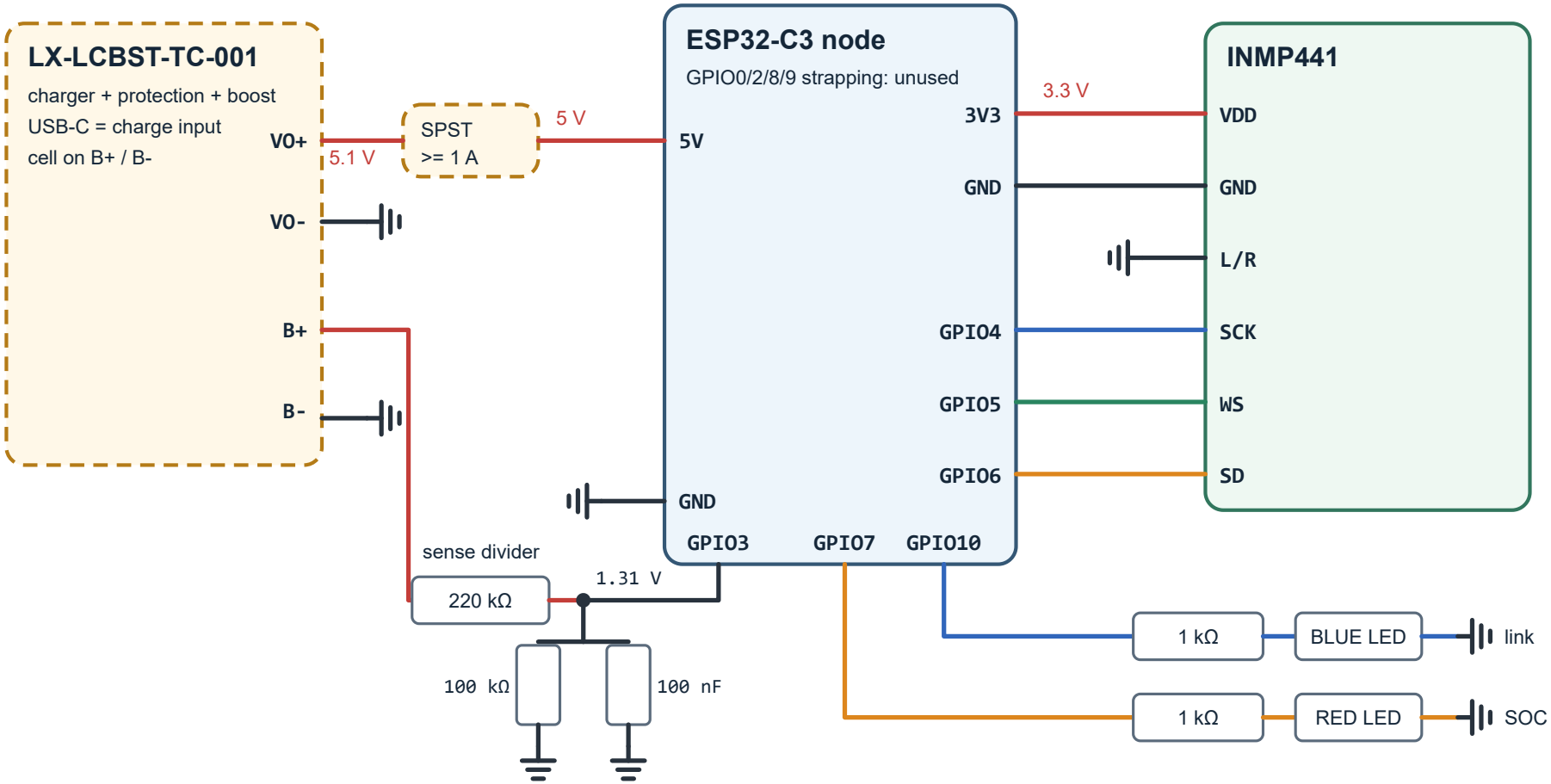
Proposed, bench-validation pending: the whole battery chain, the wearables and the LEDs. The node and hub signal wiring is the verified part.

Cells (D-81): an 18650 (3.7 V, 4.2 V full, 3000 mAh) on each node and the hub, because a node sits on a cot frame within a baby's reach and needs a cell that cannot be punctured; a 1000 mAh Li-Po pouch on the wearable, because an 18650 is 18 x 65 mm and 48 g and cannot go on a wrist. **Never LiFePO4** - the module charges to 4.2 V and an LFP cell tops out at 3.65 V. The module's ~1 A charger must come down to ~400-500 mA for the wearable's pouch (1 C is too high for it).

Runtime (D-82): the chain boosts to 5.1 V and the board drops that to 3.3 V through a *linear* regulator, which passes current rather than power - so the cell supplies roughly 1.8x the 3.3 V load current. That projects to about six hours for the wearable on a 1000 mAh pouch. These are projections, not measurements. Feeding the 3V3 pin from a 3.3 V buck-boost would roughly double it - a change to the power chain, deferred to the bench.

2 / Cot node - complete wiring (one per node, x3)

ESP32-C3 SuperMini + INMP441 + LX-LCBST + switch + battery-sense divider + 2 LEDs



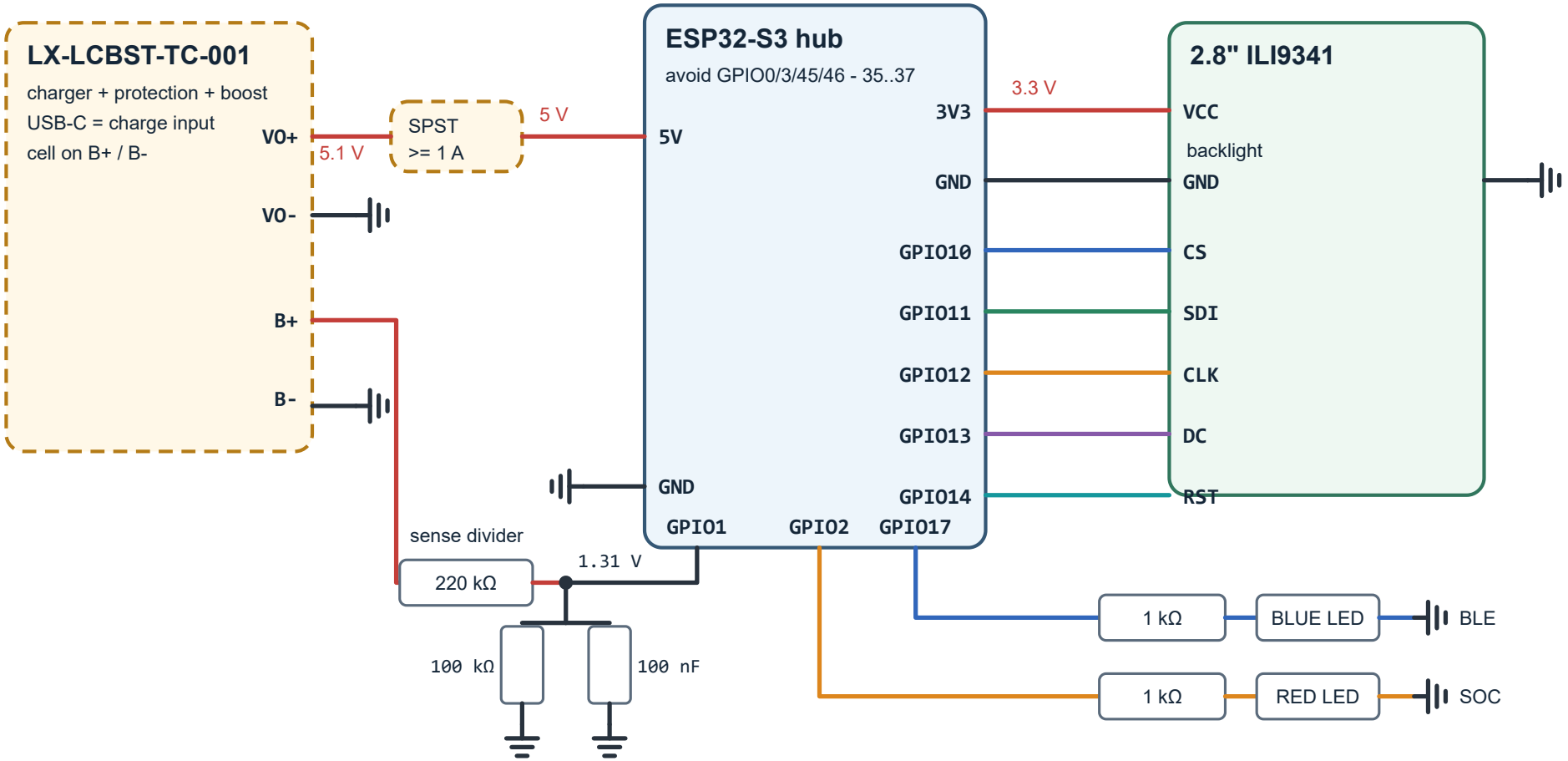
Verified on the live system: mic wiring (3V3 / GND / L/R→GND, GPIO4 SCK, GPIO5 WS, GPIO6 SD) and the node firmware. GPIO1 stays spare.

Proposed, needs bench validation: the whole battery chain, the divider and the LEDs. The divider reads about 13 uA. RED LED = SOC (off / slow / fast), BLUE LED = link (ACK-driven). Firmware for the divider and the LEDs does not exist yet - branch work.

Every ⏚ symbol is the same net: the module's VO- (battery negative through the module's protection FET). Wires join only where a dot is drawn. Pin names, not header positions. Repeat identically for nodes 2 and 3.

3 / Main hub - complete wiring (x1)

ESP32-S3-DevKitC-1 N16R8 + 2.8" ILI9341 + LX-LCBST + switch + battery-sense divider + 2 LEDs



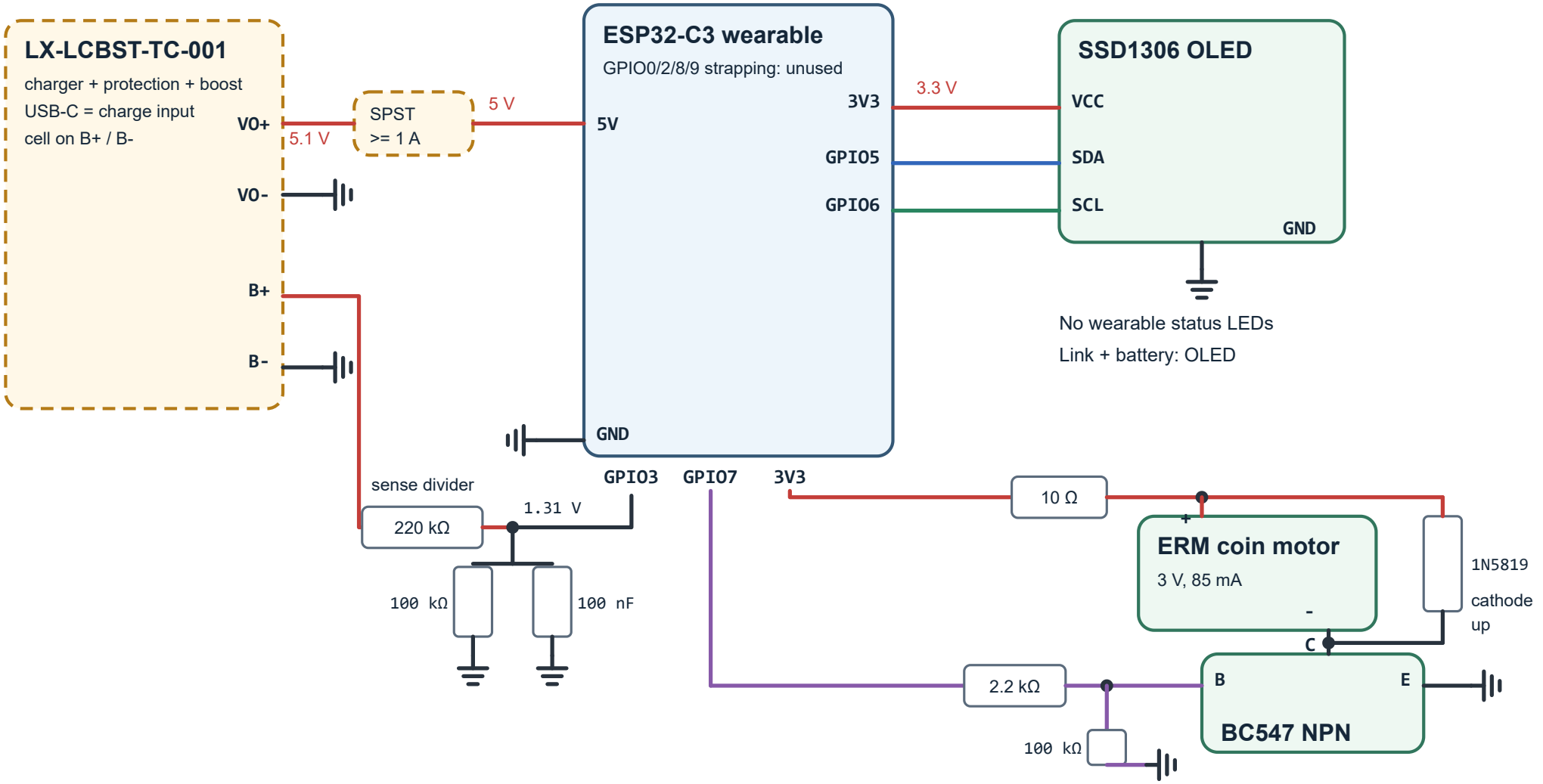
Verified on the live system: TFT signals (CS 10, MOSI 11, SCK 12, DC 13, RST 14, BL 4, VCC 3V3, GND) and the hub firmware.

Proposed, needs bench validation: the battery chain, the divider on GPIO1 (ADC1_CH0), BLUE LED on GPIO2 (BLE client connected) and RED LED on GPIO17 (SOC). The hub's peak current with TFT + radios is unmeasured - confirm the LX-LCBST holds 5.1 V during bursts before trusting it here.

GPIO35-37 are the Octal PSRAM bus - never wire them. The TFT wiring is the recorded working one for the existing module. Every ⏚ symbol is the same net: the module's VO-.

4 / Caregiver wearable - complete wiring (x1)

ESP32-C3 SuperMini + SSD1306 OLED + LX-LCBST + switch + battery-sense divider + BC547 haptic drive (status on OLED)

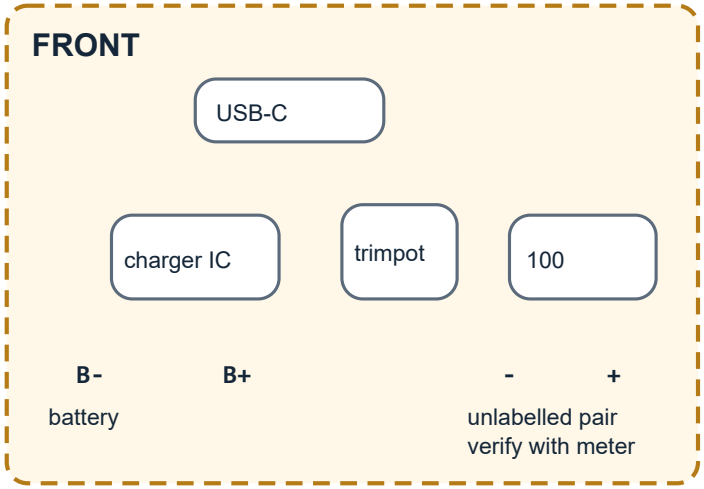


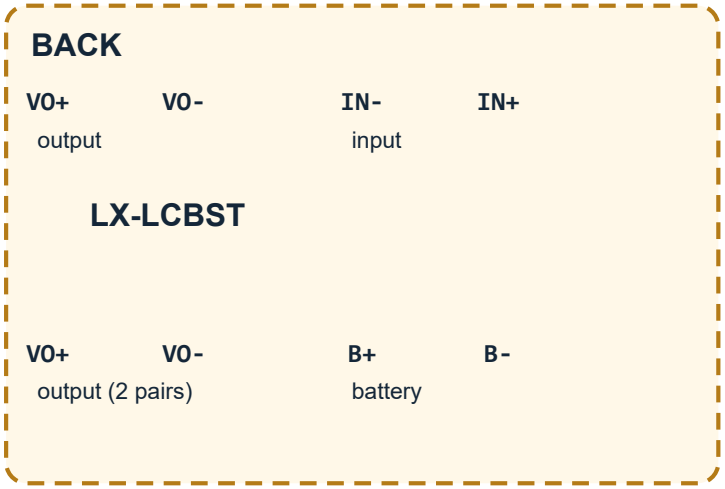
Divider confirmed: 220 kΩ from B+ and 100 kΩ to ground give 4.2 V × 100/320 = 1.31 V at the ADC (the 100 nF filters it), 13 uA of divider current; the node firmware already uses this ratio (reading × 3.2). Motor: the 10 Ω caps the stall current at 3.3 V ÷ (R + 10 Ω) - about 69 mA for a 37 Ω coin ERM, inside the BC547's 100 mA; a lower-impedance motor needs 22 Ω or a 2N2222A. GPIO7 drives the switch in 250 ms pulses. No LEDs are fitted: link status is shown on the OLED; battery display needs the divider and firmware support.

Every ⏚ symbol is the same net; wires join only where a dot is drawn. The BC547 emitter is the motor's ground, and the flyback sits across the motor with the cathode up.

5 / LX-LCBST-TC-001 - pad map and bench procedure

From the user's front-and-back photograph - Chinese silk text translated





Bench procedure, before any ESP32 is connected:

1 - Cell on B+/B-: turn the trimpot until VO+/VO- reads 5.1 V on a meter, no load.

2 - USB-C in: measure the charge current with a series meter - the cell must allow it.

3 - Confirm the charge ends at 4.2 V.

4 - Switch off: confirm the load is dead and measure the battery drain (boost quiescent unknown).

5 - Run one node and compare it with the USB baseline.

No T pad on this board - the listing's T-shorting notes belong to other variants. Never short anything.

Unknown until measured: the actual charge current (the program resistor is unreadable in the photo), the boost quiescent drain when off, the protection thresholds, and what the front - / + pair connects to. Buy ONE module for validation; only then buy the remaining three (plus the wearable's, pending the same check).